

Based on the provided lab manual, here are the main observations on how speech is produced.

The Source-Filter Model in Practice

The document shows how the source-filter model of speech production works. The "time-varying excitation" serves as the source. This can be periodic, like the vibrations of vocal folds for vowels (/i/, /a/), or aperiodic, like the turbulent noise from a constriction for a fricative (/h/) or the burst of sound from a plosive (/k/).

The "time-varying system" is the filter, which is the vocal tract itself. The shape of this filter changes for each sound based on the position of the articulators, like the tongue and lips. For example, to produce the vowel /i/, the tongue is high and front, creating a narrow opening in the vocal tract. In contrast, for /a/, the tongue is low and central, resulting in a wider opening. The final speech sound is a mix of the source and filter.

Acoustic Cues and Phonetic Events

There is a clear connection between how a sound is physically made (phonetics) and its features in a waveform or spectrogram (acoustics).

Voicing: Voiced sounds, like vowels (/i/, /E/) and nasals (/m/), occur when the vocal folds vibrate. This is visible in the spectrogram as distinct vertical lines, known as pitch harmonics.

Manner of Articulation: The manner of articulation creates unique acoustic patterns.

Plosives like /k/, /t/, and /p/ involve a complete blockage of airflow followed by a sudden release. This is shown in the spectrogram as a "silence bar" (white space) followed by a burst of energy, which is stronger at high frequencies for unvoiced stops like /k/ and /t/.

Vowels have a clear formant structure—dark, horizontal bands that represent the resonances of the vocal tract. The frequencies of these formants help distinguish one vowel from another.

Fricatives like /h/ are made by creating a narrow opening, resulting in a turbulent, noisy sound. This appears in the spectrogram as random energy without clear pitch harmonics.

Place of Articulation: The location of the constriction (e.g., velar for /k/, dental for /t/) also affects the sound produced, especially in the frequency distribution of the burst in plosives.

The Spectrogram as a Rich Visualization Tool

The spectrogram is a useful tool because it shows how the frequency and energy of a speech signal change over time.

Formant Dynamics: It clearly displays the formant structures that define vowels. The movement of these formants illustrates the dynamic nature of speech. For the diphthong /ai/, the spectrogram shows a transition from the formant structure of /a/ to that of /i/, highlighting the continuous movement of the tongue.

Coarticulation: The spectrogram reveals how nearby sounds affect one another, a phenomenon known as coarticulation. For instance, the analysis notes that the formant structure for the nasal sound /m/ is influenced by the vowel /E/ that follows it. Similarly, traces of formants are seen in the fricative /h/ because of the influence from the following diphthong /ai/. This illustrates that sounds are not produced in isolation.

In summary, the document shows that a speech signal is highly structured. Its acoustic properties, as displayed in a spectrogram, offer a detailed view of the dynamic and physical processes involved in how humans produce speech.